Modeling Household Electricity Price Volatility with EGARCH and GARCH, Local Projections, and Random Forest

1. Introduction

Europe's Energy Crisis can be traced back to Russia's invasion of Ukraine in 2022 in which the prices of oil, coal, and gas increased by about 40%, 130%, and 180% respectively because of Europe's once immense dependence on energy imports from Russia. The energy sector as a whole is especially vulnerable to geopolitical shocks because of each country's reliance on these resources; while the war in Ukraine has inspired policy that seeks to shift away from fossil fuels to renewable energy sources, these goals have largely not come to fruition. This paper seeks to model renewable energy price volatility under geopolitical uncertainty in the post-2021 period. More specifically, how does global economic policy uncertainty as measured by the Global EPU Index impact European household electricity price volatility? Additionally, what are the leading macroeconomic indicators in predicting volatility regimes? Previous literature that has utilized uncertainty indexes have focused on different markets like that of commodities, as well as stock returns for renewable energy assets (Ozdemir 2025, Kim 2026). This paper on the other hand focuses solely on household electricity prices which are also viewed through the lens of volatility regimes. In addition to the dependent variable of EU household electricity prices and the EPU index as an exogenous variable, EU Natural Gas Price and Brent Crude Oil price serve as energy market controls in the Random Forest model. Euro area HICP inflation and German industrial production are used as the macroeconomic controls; through the use of volatility models like GARCH and EGARCH, Local Projections, and Random Forest machine learning, this paper ultimately found that there is near-universal volatility persistence across EU member states, as well as significant asymmetric leverage effects in 14 out of the 27 EU countries.

Additionally, EU industrial production and the EPU index were the two most important predictors in identifying high volatility regimes, and these findings can help build a framework for anticipating future geopolitical shocks.

2. Literature Review

Engle's 1982 paper is widely recognized as the first academic study to introduce the autoregressive conditional heteroskedasticity (ARCH) model. Engle seeks to answer the question of whether the uncertainty of a time series can change systematically over time, and if it can be predicted using past information. Engle specifically uses monthly U.K. inflation data which he attempts to forecast the uncertainty of. Engle's most important findings include the presence of volatility clustering in numerous economic time series, as well as the ARCH model's ability to forecast conditional variance. This paper is foundational not only for this study, but for any paper that seeks to model volatility, whether it's through the use of the GARCH (Bollerslev 1986) or EGARCH models (Nelson 1991). The conditional heteroskedasticity seen in energy price returns motivates the implementation of volatility models, which can be determined through Engle's ARCH LM Test.

Jorda's paper titled "Estimation and Inference of Impulse Responses by Local Projections" offers a viable alternative to vector autoregressions (VAR). Jorda argues that VARs are often misspecified in the data generating process, and that local projections have numerous advantages: "they can be estimated by simple least squares; they are robust to misspecification of the DGP; and they easily accommodate experimentation with highly nonlinear specifications that are often impractical or infeasible in a multivariate context" (Jorda 2005, 162). Jorda specifically utilizes Monte Carlo simulations in order to evaluate local projections, and one experiment finds

that “local projections deliver impulse responses that are robust to lag length misspecification, consistent, and only mildly inefficient relative to the responses from the true DGP” (Jorda 2005, 168). Local projection models are useful when analyzing renewable energy price volatility because they can assess causality with the use of impulse response functions. Additionally, LPs can also estimate responses at specific horizons and indicate their overall significance, which is useful when analyzing supply and demand dynamics of energy price volatility.

Chowdhury’s article titled “Geopolitical risk and energy market tail forecasting: An explainable machine learning approach” is motivated by volatility in energy markets as a result of geopolitical risks. More specifically, Chowdhury’s article asks the following question: how do geopolitical risks influence energy market tail risk forecasting? Tail risk refers to the risk of extreme price movements and given how dependent countries are on energy markets, it’s important to find suitable forecasting methods in the midst of negative shocks. The S&P Goldman Sachs Commodity Index (GSCI) Energy spot price index is used with data covering January 1, 2000 to March 26, 2024. Chowdhury argues that the commonly used ARIMA and GARCH methods are too limited in scope to be used in predicting energy prices (Chowdhury 2025 , 2). Chowdhury therefore utilizes CAViAR for the estimation of energy market tail risk as well as machine learning models like Random Forest in forecasting. The paper finds that Light Gradient Boosting Machine consistently exhibited the lowest MAE, MAPE, and MSE values. For the purposes of this paper, Random Forest remains a suitable machine learning method because on the training set, RF had the second lowest MAE and MAPE (Chowdhury 2025, 11). Random Forest may have some advantages over LGBM if it exhibits less overfitting and is less sensitive to outliers, which is important when modeling shocks that are volatile in nature; this is especially true for modeling energy price volatility. The Random Forest model may also prove to

be more suitable than certain GARCH methods, and this paper utilizes both models in order to assess their applicability to energy price volatility.

Ozdemir's article titled "Volatility Modeling of the Impact of Geopolitical Risk On Commodity Markets" focuses specifically on commodity futures returns, like those of precious and industrial metals, agricultural products, and energy products. The paper seeks to answer the question of what the primary cause of price fluctuations are, and whether these fluctuations are a result of the event itself or the actions that follow these geopolitical shocks. The paper uses data from January 4th 2010 to June 30th 2023 and uses the GPR, GPRACT, and GPRTHREAT indices in order to examine their effects on commodity market volatility. The paper utilizes the EGARCH model in order to allow for asymmetric effects: the paper finds that the "presence of the leverage effect" is "in most of the commodities analyzed" (Ozdemir 27, 2025). The findings in this study are important for the goals of this paper because it uses two distinct indices to differentiate between geopolitical threat actions and geopolitical events. Additionally, the paper found that negative shocks have a particularly pronounced effect on energy markets given the global dependence on these products.

Kim's paper titled "Predicting renewable energy stock volatility: A GARCH-CNN approach with indicator analysis" seeks to answer the question on how to forecast renewable energy stocks, and how to potentially predict their volatility dynamics considering that these assets have grown in recent years and the money of investors is at stake. Kim therefore utilizes both the GARCH and Convolutional Neural Network (CNN) in order to predict renewable energy stock volatility, as well as three groups of indicators which includes internal stock, financial market, and policy uncertainty as inputs. The study ultimately finds that the GARCH-CNN model outperforms the traditional GARCH model, but this is heavily dependent on the

indicators that are used. Additionally, the inclusion of policy uncertainty indicators reduces overall forecasting performance. While Kim's study focuses specifically on renewable energy assets rather than household electricity prices, the methodology utilized in this paper provides a suitable framework in modeling volatility. While the GARCH model has its limitations, it can provide useful economic information on energy price volatility when paired with other models. This paper also utilizes the EPU index which is a policy uncertainty index, which can potentially worsen predictive accuracy.

Ganczarek-Gamrot in their paper "Electricity Price Volatility and the Performance of Machine Learning Forecasting Models in European Energy Markets" seeks to answer the question of how market volatility influences the accuracy of ML-based forecasting approaches (Ganczarek-Gamrot 2025, 2). The paper mainly utilizes ENTOS-E daily data from 2023-2024 in order to analyze electricity price dynamics across seven European bidding zones. The paper found that for all bidding zones, the random forest model exhibited the best RMSE and NMAE (Ganczarek-Gamrot 18). Additionally, "For highly volatile markets…the best forecast results were obtained for models trained on short or very short periods preceding the forecast," and "For markets with lower price volatility…the best results were obtained with larger time windows" (Ganczarek-Gamrot 2025, 18). The findings from this study are crucial for this paper because they indicate that highly volatile energy markets are best forecasted using only recent data. This therefore supports the use of short training windows, whereas lower volatile energy markets may perform better with larger training windows.

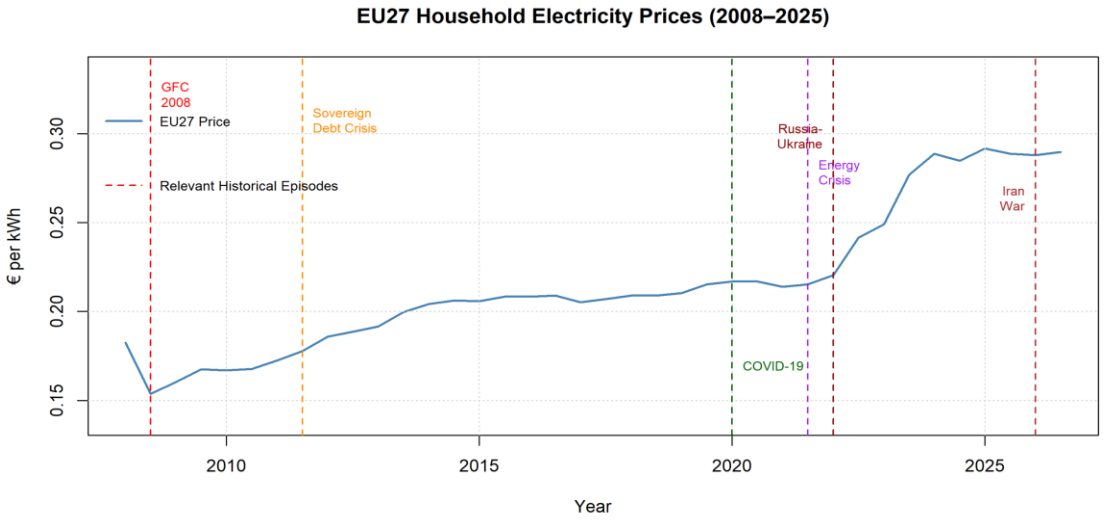
This paper is uniquely positioned within the traditional energy markets literature because of its focus on semi-annual household electricity price data and using the Random Forest model to predict high volatility regimes. Traditionally, Random Forest models in this discipline have

sought to analyze how volatility impacts the accuracy of machine learning models rather than using them to predict high volatile regimes (Ganczarek-Gamrot 2025). Additionally, the focus of these studies has mainly been on commodity markets as well as energy asset returns (Ozdemir 2025, Kim 2026). This study seeks to combine volatility models, Local Projections, and machine learning models in order to assess asymmetric effects across specific EU member states, and to determine the most significant indicators in predicting volatility regimes.

3. Data

The main variable of interest in this study is EU Electricity Prices for Household Consumers obtained from Eurostat. This data tracks household electricity prices semi-annually, and it is broken down by consumption band and household size. This paper utilizes data from Band DC, which is defined as annual household consumption between 2500 and 5000 kilowatts and is the most widely used band in traditional literature on European household electricity prices. The full sample begins in 2008 which is used for the time series plots and stationarity tests. Given the fact that this data is recorded semi-annually and therefore results in a limited number of observations, this paper utilizes the panel approach for volatility modeling and random forest, which results in a total of 972 observations for the 27 EU countries in the sample and a total of 36 periods. Additionally, the post-2021 crisis subsample which runs from 2021 to 2025 is the main subsample for analysis. The Global Economic Policy Uncertainty Index is used as the primary uncertainty measure, which is constructed from newspaper coverage across 21 major countries. These values are recorded monthly but for the purposes of this paper, the values are aggregated to semi-annual averages in order to match the frequency of the Eurostat data. The EPU index is utilized as the impulse variable in the Local Projections analysis, but it is also utilized as a

variable in the Random Forest model. The machine learning model also includes a number of controls: EU natural gas price, Brent crude oil price, Euro area HICP inflation, and German industrial production. These monthly values were similarly converted to semi-annual averages in order to match the frequency of the Eurostat data. German industrial production serves as a proxy for EU industrial production given that its production accounts for about 25% of EU total industrial output. All datasets, except for the Eurostat household electricity price data, were sourced from FRED.



The 2007 Recession, the European Sovereign Debt Crisis, the Russia-Ukraine War, COVID-19 shock, European Energy Crisis, and Iran War are cited as relevant historical episodes. This graphs paints a clear picture where there was a significant decline in European household electricity prices during the global financial recession, followed by a gradual rise in prices. Prices rises substantially after the European Energy Crisis and appear to stabilize during the Iranian conflict.

Summary Statistics: EU27 Household Electricity Prices

Statistic	Price Level	Full Sample Returns	Pre-Crisis (2008–2020)	Post-Crisis (2021–2025)
Mean	0.215621	0.012465	0.006886	0.024089
Standard Deviation	0.039912	0.040645	0.040830	0.039384
Minimum	0.153700	-0.172295	-0.172295	-0.013952
Maximum	0.291600	0.105842	0.044574	0.105842
Skewness	0.702836	-2.066599	-3.467954	1.062568
Excess Kurtosis	-0.432056	10.474386	13.221021	-0.025787

The summary statistics table reports figures for EU27 aggregate price level: the price level mean is 0.2156 with a standard deviation of 0.039912 which represents an upward trend in prices across the sample period due to the shock of the European energy crisis. The price level distribution is positively skewed and playkurtic, while the full sample log-returns are negatively skewed and exhibit extreme kurtosis.

Stationarity Tests: EU27 Household Electricity Prices

Series	Test	Statistic	p-value	H_0	Decision
Price Level					
EU27 Price (Level)	ADF (k=3)	-2.6722	0.311	Unit Root	Non-Stationary
EU27 Price (Level)	PP (k=3)	-9.8036	0.508	Unit Root	Non-Stationary
EU27 Price (Level)	KPSS	0.9079	0.010	Stationary	Non-Stationary
First Difference (Δ)					
EU27 Price (Δ)	ADF (k=3)	-2.6771	0.309	Unit Root	Non-Stationary
EU27 Price (Δ)	ADF (k=1)	-3.2660	0.092	Unit Root	Stationary*
EU27 Price (Δ)	ADF (k=0)	-6.4574	0.010	Unit Root	Stationary
EU27 Price (Δ)	PP (k=3)	-36.2104	0.010	Unit Root	Stationary
EU27 Price (Δ)	KPSS	0.2039	0.100	Stationary	Stationary

Before first-differencing, the EU27 price series is non-stationary. The first-difference results indicate that the PP test rejects the unit root and the KPSS test fails to reject the null hypothesis, but the ADF test fails to reject the unit root. The ADF test in particular has lower

power in small samples and may fail to reject the null hypothesis even when the series is stationary. With a lag order of 1, the ADF test rejects the null hypothesis at 10% and with a lag order of 0, the ADF test rejects the null hypothesis at 1%. With these results, the differenced series is I(1).

4. Empirical Methodology

This paper utilizes a number of methods that have not been widely used in conjunction with one another, especially when analyzing household electricity prices. These methods include GARCH and EGARCH models to capture volatility, local projections to assess causality through impulse response functions, and the random forest model to predict volatility regimes.

4.1. Volatility Modeling

The LM (Lagrange Multiplier) test for the Autoregressive Conditional Heteroskedasticity (ARCH) effect was first explored by Engle in 1982. Specifically, it estimates a suitable model and obtains errors from the regression model as well as the square of errors. The ARCH model is defined by the equation below:

$$Y_t = \delta + \phi_1 Y_{t-1} + \varepsilon_t$$

$$\varepsilon_t^2 = h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 \cdots + \alpha_1 \varepsilon_{t-q}^2$$

While this paper doesn't estimate an ARCH(q) model, Engle's 1982 paper serves as a foundational study on volatility modeling, and because the GARCH and EGARCH models serve as natural extensions of the ARCH model. Additionally, the ARCH-LM test is used in order to

measure autoregressive conditional heteroskedasticity in order to support the use of volatility models.

GARCH, which stands for Generalized Autoregressive Conditional Heteroskedasticity, serves as an extension to Engle's ARCH model (Bollerslev 1986). The GARCH model adds lags of the variance, ht-1 and is defined by the equation below:

$$Y_t = \delta + \phi_1 Y_{t-1} + \varepsilon_t$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}$$

For the purposes of this paper, which utilizes GARCH(1,1), Y_t measures the log return of electricity price.

EGARCH extends Bollerslev's GARCH model in two ways: first, non-negativity constraints on parameters are automatically satisfied and it helps to capture the leverage effect (Nelson 1991). The leverage effect is defined as negative shocks increasing volatility more than positive shocks of the same magnitude. The EGARCH(p,q) variance equation, which is specified in log form, is shown below:

$$\ln h_t = \omega + \sum_{j=1}^{q} \left[\alpha_j \left(\left| \frac{\varepsilon_{t-j}}{\sqrt{h_{t-j}}} \right| - \sqrt{\frac{2}{\pi}} \right) + \gamma_j \frac{\varepsilon_{t-j}}{\sqrt{h_{t-j}}} \right] + \sum_{i=1}^{p} \beta_i \ln h_{t-i}$$

Since this paper utilizes EGARCH(1,1), the equation is given below:

$$\ln \sigma_t^2 = \omega + \alpha_1 \left(\left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right| - \sqrt{\frac{2}{\pi}} \right) + \gamma_1 \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \beta_1 \ln \sigma_{t-1}^2$$

EGARCH(1,1) is especially useful for the purposes of this paper because of the asymmetry coefficient which can measure the leverage effect across the 27 EU member states.

4.2. Local Projections

The panel LP at horizon h as defined by Jorda (2005) is given by the equation below:

$$r_{i,t+h} = \alpha_i + \beta_h \Delta \ln(\mathrm{EPU}_t) + \gamma_1 r_{i,t-1} + \gamma_2 r_{i,t-2} + \varepsilon_{i,t+h}^{(h)}$$

Where $r_{i,t+h}$ is the semi-annual log return for country i at horizon h, α_i denotes country fixed effects, $\beta_h$is the LP impulse response coefficient at horizon h, $\Delta\ln(\mathrm{EPU}_t)$is the first difference of the log Global EPU index, $\gamma_1$and $\gamma_2$are coefficients on lagged returns included as controls, and $\varepsilon_{i,t+h}^{(h)}$is the regression error. The sequence $\{\hat{\beta}_h\}_{h=0}^{H}$traces the impulse response function estimated separately at each horizon.

IRF confidence intervals can be constructed in three different ways: residual bootstrapping, the Bayesian method, and the Delta method. This paper utilizes the Delta method with Newey West standard errors as well as residual bootstrapping with 500 replications for the VAR comparison. The advantage of residual bootstrapping is that it is nonparametric, but it may draw near-explosive samples. The Delta method on the other hand derives asymptotic variance of IRF through the chain rule.

Local projections do not require the same structural restrictions needed by the SVAR model since it estimates reduced-form projections directly. This means that the identifying assumption is that the natural log of EPU is predetermined with respect to country-level electricity price returns conditional on lagged returns and country fixed effects.

4.3. Random Forest

The output is the mean prediction of individual trees, which is represented by the equation below:

$$Y = \frac{1}{B}\sum_{b=1}^{B} T_b(X)$$

Since this study utilizes the random forest model to predict volatility regimes, the appropriate equation is presented below:

$$\widehat{R_{i,t}} = \backslash undersetc \in \{\text{High, Low}\}\text{argmax}\frac{1}{B}\sum_{b=1}^{B} \mathbb{1}\left[T_b\left(x_{i,t}\right) = c\right]$$

where $\hat{R}_{i,t}$ is the predicted volatility regime for country i at time t, $B = 500$ is the number of classification trees, and $\mathbf{x}_{i,t}$is the vector comprising the EPU shock, two lags of log returns, and semi-annual changes in EU natural gas prices, Brent crude oil, Euro area CPI, and German industrial production (meant to represent EU industrial production). The Random Forest model utilizes a train-test split where the training set represents the period between 2007 and 2020 (the pre-crisis period), and the test set which represents the post-crisis period (2021-2025). The training set is downsampled given the imbalance in observations. Rolling window cross-validation is also utilized with the Random Forest model in order to evaluate performance.

5. Results

Engle's LM ARCH Test: Country-Level Log Returns

Country	LM Statistic	p-value	ARCH Effects
AT	**25.3032**	**0.0001**	**Yes (5%)**
BE	3.2677	0.6588	No
BG	1.1150	0.9528	No
CY	0.8995	0.9703	No
CZ	3.0007	0.6999	No
DE	0.6631	0.9849	No
DK	6.2602	0.2817	No
EE	5.9108	0.3150	No
EL	2.5738	0.7653	No
ES	2.5072	0.7754	No
FI	0.1941	0.9992	No
FR	2.4807	0.7794	No
HR	2.8189	0.7279	No
HU	0.7043	0.9827	No
IE	**21.3871**	**0.0007**	**Yes (5%)**
IT	2.0759	0.8386	No
LT	1.5368	0.9088	No
LU	0.8437	0.9742	No
LV	2.3960	0.7921	No
MT	8.2824	0.1413	No
NL	**16.1435**	**0.0064**	**Yes (5%)**
PL	8.9967	0.1092	No
PT	1.9546	0.8554	No
RO	10.8221	0.0550	Yes (10%)
SE	8.4402	0.1336	No
SI	3.1373	0.6788	No
SK	5.4290	0.3658	No

At the EU27 aggregate level, there are no statistically significant ARCH effects in the log returns for household electricity price. Averaging essentially obscures country-level volatility since the spikes offset each other. According to these results, 3 out of 27 countries show significant ARCH effects at the 5% level and only 1 country shows significant ARCH effects at the 10% level. Additionally, ARCH tests are limited in small samples, and GARCH and EGARCH by comparison can be used for low frequency volatility.

GARCH(1,1) Estimates: EU27 Country-by-Country

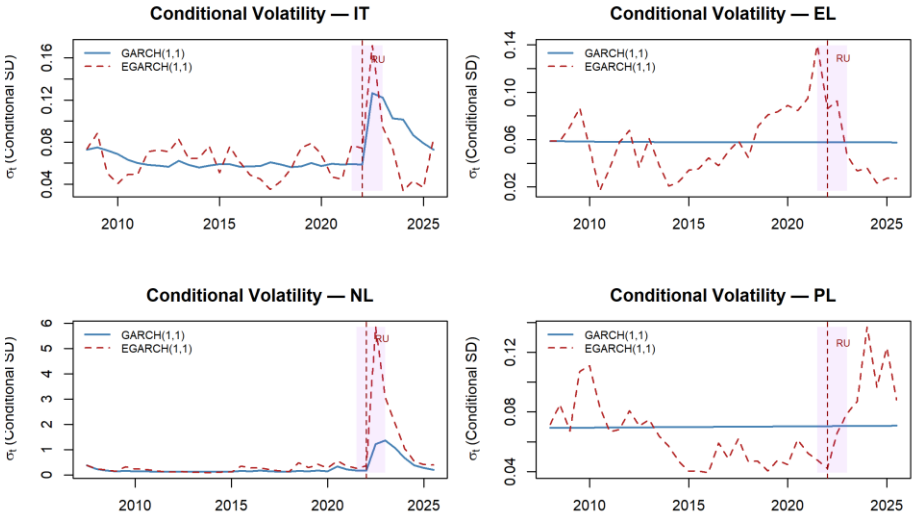
Country	ω (omega)	α_1 (alpha)	β_1 (beta)	ν (shape)	α+β (persist.)	LR Volatility	AIC	BIC
AT	0.000136	0.471535	0.527465	8.7168	0.999000	0.368818	-3.6166	-3.3967
BE	0.000042	0.000000	0.999000	5.9138	0.999000	0.204420	-1.9843	-1.7643
BG	0.000000	0.000012	0.981702	27.1528	0.981714	0.000051	-3.2288	-3.0089
CY	0.000000	0.000000	0.997874	99.9995	0.997874	0.008201	-1.1204	-0.9005
CZ	0.000421	0.000000	0.999000	2.1000	0.999000	0.648849	-2.2676	-2.0499
DE	0.000221	0.000000	0.999000	2.1739	0.999000	0.470125	-3.8452	-3.6275
DK	0.000545	0.729764	0.269236	3.7910	0.999000	0.738007	-2.6911	-2.4711
EE	0.002345	0.000000	0.997884	2.1000	0.997884	1.052675	-2.1257	-1.9058
EL	0.000370	0.000000	0.888372	24.5861	0.888372	0.057609	-2.5557	-2.3358
ES	0.000192	0.000000	0.999000	2.6407	0.999000	0.437895	-2.2777	-2.0577
FI	0.000398	0.113849	0.719690	2.5987	0.833539	0.048918	-3.5072	-3.2872
FR	0.000306	0.282715	0.591962	13.0226	0.874677	0.049424	-3.4528	-3.2328
HR	0.000069	0.000000	0.956555	2.4512	0.956555	0.039964	-3.4133	-3.1956
HU	0.000345	0.389281	0.567879	3.4690	0.957160	0.089770	-2.6322	-2.4122
IE	0.000734	0.589051	0.409949	99.7757	0.999000	0.856712	-1.8340	-1.6140
IT	0.000994	0.172390	0.622504	4.8819	0.794894	0.069601	-2.3706	-2.1485
LT	0.002112	0.000000	0.999000	2.1000	0.999000	1.453129	-1.9472	-1.7273
LU	0.000000	0.000001	0.962080	2.3190	0.962081	0.000011	-3.5276	-3.3099
LV	0.000319	0.000000	0.999000	2.1909	0.999000	0.565131	-1.7125	-1.4926
MT	0.000416	0.999000	0.000000	3.1677	0.999000	0.644915	-3.4743	-3.2544
NL	0.013415	0.700153	0.298847	2.2041	0.999000	3.662685	-0.9495	-0.7318
PL	0.000011	0.000000	0.998999	7.3231	0.998999	0.103712	-2.2392	-2.0192
PT	0.000037	0.000000	0.999000	3.4038	0.999000	0.192808	-3.1732	-2.9532
RO	0.010955	0.999000	0.000000	2.2562	0.999000	3.309294	-1.7905	-1.5706
SE	0.000652	0.435582	0.563248	4.9248	0.998830	0.746490	-2.3290	-2.1113
SI	0.000282	0.292376	0.706624	2.5445	0.999000	0.531201	-2.6121	-2.3922
SK	0.000081	0.000000	0.972106	99.9970	0.972107	0.053812	-2.9348	-2.7148

According to the GARCH(1,1) results, unit-root persistence is dominant across the table because of the limited sample size. There are some important findings to address: Italy has the lowest persistence at about 0.794894, as well as the fastest mean reversion. Long-run volatility also varies from country to country which indicates that there are country-level differences in price level volatility. The most stark differences are between Romania and Luxembourg, which have long-run volatility figures of 3.309294 and 0.000011 respectively. These differences in long-run volatility may be explained through varying country-level policy as well as different energy mixes.

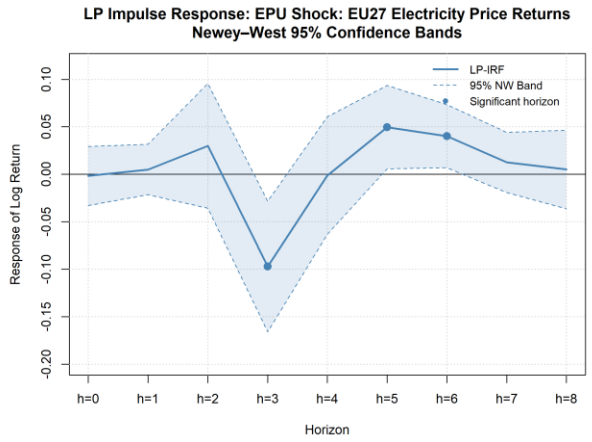
EGARCH(1,1) Estimates: EU27 Country-by-Country

Country	ω (omega)	α_1 (alpha)	γ_1 (gamma)	β_1 (beta)	ν (shape)	Leverage Effect	AIC	BIC
AT	0.068418	0.337326	0.833550	1.000000	99.9995	No	-3.6559	-3.3920
BE	-1.962381	0.376472	0.460206	0.643897	100.0000	No	-2.1665	-1.9026
BG	-3.604900	-0.913319	0.521007	0.451785	99.4520	No	-3.3918	-3.1279
CY	**-1.471550**	**0.420431**	**-1.677718**	**0.666685**	**11.0500**	**Yes (sig.)**	**-1.3055**	**-1.0416**
CZ	-1.440581	0.969551	0.805350	0.737017	2.2374	No	-2.3699	-2.1087
DE	**-0.212431**	**-0.243970**	**-1.251215**	**0.977318**	**3.1149**	**Yes (sig.)**	**-4.0573**	**-3.7961**
DK	-1.208708	0.305684	0.853949	0.801480	3.8264	No	-2.7212	-2.4573
EE	**-0.809352**	**1.437794**	**-0.936318**	**0.757563**	**2.1000**	**Yes (sig.)**	**-2.1448**	**-1.8808**
EL	**-1.317545**	**-0.246058**	**-1.394014**	**0.776720**	**7.9327**	**Yes (sig.)**	**-2.8488**	**-2.5848**
ES	**-0.998668**	**-0.530791**	**-1.690687**	**0.840307**	**3.4831**	**Yes (sig.)**	**-2.5654**	**-2.3015**
FI	-1.446370	0.720895	-0.198349	0.801062	2.8303	Yes (insig.)	-3.5352	-3.2713
FR	**-0.714288**	**0.455735**	**-1.175305**	**0.901513**	**11.8259**	**Yes (sig.)**	**-3.6690**	**-3.4051**
HR	**-1.025133**	**0.292293**	**-1.192032**	**0.833725**	**3.2163**	**Yes (sig.)**	**-3.7866**	**-3.5254**
HU	-2.129290	-0.304346	0.840770	0.643473	4.5291	No	-2.5719	-2.3080
IE	-0.635868	0.389445	0.927673	0.868166	100.0000	No	-1.8189	-1.5550
IT	-4.621002	0.652215	-0.117773	0.194689	7.5588	Yes (insig.)	-2.3983	-2.1317
LT	**-3.156452**	**8.320637**	**-4.233232**	**0.384418**	**2.1000**	**Yes (sig.)**	**-2.4296**	**-2.1656**
LU	**-2.057091**	**-1.393621**	**-1.615335**	**0.676078**	**2.3133**	**Yes (sig.)**	**-4.4938**	**-4.2326**
LV	**-0.105371**	**-0.322162**	**-0.916923**	**0.960024**	**5.0806**	**Yes (sig.)**	**-2.3291**	**-2.0652**
MT	-5.117043	2.026830	5.065028	-0.016122	2.1750	No	-3.6975	-3.4335
NL	-0.566994	0.910832	2.518306	0.769256	2.1000	No	-1.1082	-0.8470
PL	-1.079765	-0.028066	0.733077	0.805344	99.9999	No	-2.2903	-2.0264
PT	**-0.508723**	**-0.106853**	**-1.019163**	**0.914375**	**6.7041**	**Yes (sig.)**	**-3.6283**	**-3.3644**
RO	**-0.733579**	**1.634241**	**-2.243529**	**0.913316**	**2.6212**	**Yes (sig.)**	**-2.2227**	**-1.9588**
SE	-1.265829	0.799985	0.081791	0.765735	70.0576	No	-2.3006	-2.0394
SI	**-0.023045**	**-1.174243**	**-1.231188**	**0.989996**	**3.1130**	**Yes (sig.)**	**-3.0290**	**-2.7651**
SK	**-1.315661**	**-0.868290**	**-1.524100**	**0.798638**	**4.7150**	**Yes (sig.)**	**-3.3324**	**-3.0685**

According to the EGARCH(1,1) results, 14 out of 27 countries show significant leverage effects which indicates that negative shocks increase electricity price volatility more than positive shocks of the same magnitude. Notable asymmetry coefficients to point out are the results for Lithuania and Romania, which have asymmetry coefficients of -4.233232 and -2.243529 respectively. There are two EU member states that have insignificant leverage effects: Liechtenstein and Italy.



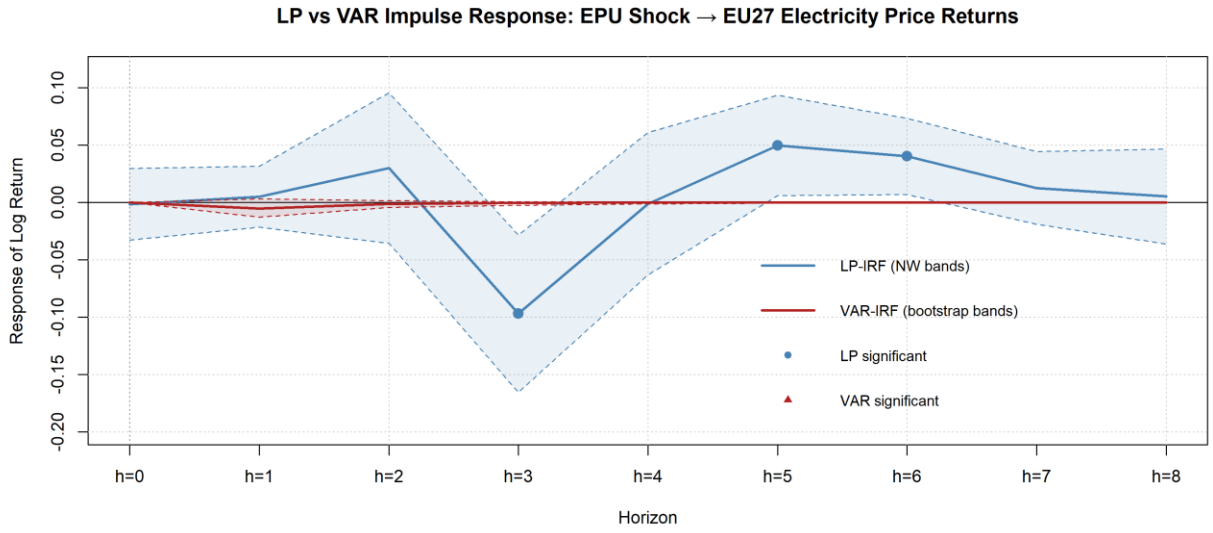
The following countries were selected for the conditional volatility plots: Italy, Greece, the Netherlands, and Poland. Italy exhibited an insignificant leverage effect, while the asymmetry coefficient for Greece was significant. EGARCH did not identify a leverage effect for the Netherlands, but the EU member had the highest long-run volatility. There were no observable leverage effects or high long-run volatility for Poland which makes it a viable plot for comparison. Italy in particular shows meaningful GARCH time-variation because its persistence figure was below 0.90.



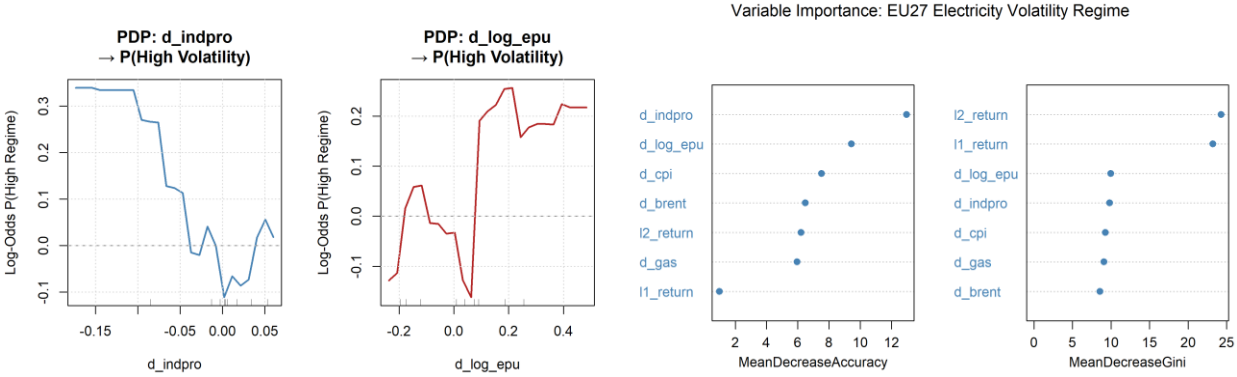
LP Impulse Responses: EPU Shock to EU27 Electricity Price Log Returns

Horizon	Estimate	SE (NW)	95% CI Lower	95% CI Upper	Significant
0	-0.001677	0.015901	-0.032843	0.029490	No
1	0.004994	0.013556	-0.021576	0.031564	No
2	0.029895	0.033492	-0.035749	0.095538	No
3	**-0.097114**	**0.035056**	**-0.165823**	**-0.028405**	**Yes ***
4	-0.001246	0.031633	-0.063246	0.060754	No
5	**0.049669**	**0.022358**	**0.005848**	**0.093491**	**Yes ***
6	**0.040123**	**0.017006**	**0.006791**	**0.073455**	**Yes ***
7	0.012556	0.016153	-0.019104	0.044216	No
8	0.005183	0.021098	-0.036170	0.046536	No

The LP impulse response plot from horizons 0 to 2 don't show a meaningful impact of EPU shocks on electricity price returns because retail household prices adjust slowly due to regulated tariffs and long-term supply contracts. Horizon 3 represents a significant negative response which indicates that an EPU shock reduces log returns. This response appears counterintuitive, but an EPU shock may push electricity prices down because of initial reduced energy demand. Later horizons such as horizons 5 and 6 represent a significant positive response which indicates that an EPU shock increases electricity price log returns. This is an expected finding since price increases are prevalent during energy crises.



The LP and VAR overlay plot reveal that the Local Projections model is more suitable for datasets with small samples than the VAR model (Jorda 2005). The VAR IRF is largely flat across all horizons since it was estimated using the EU27 aggregate data. The LP IRF on the other hand distinctly shows which horizons had significant positive or negative responses as a result of the impulse variable. Ultimately, the LP model reveals that global uncertainty has a significant effect on EU household electricity prices at specific horizons, specifically for horizons 3, 5, and 6.



The two most important predictors of EU electricity volatility regimes is industrial production change and the log of the EPU index. When European industrial production falls, then electricity demand similarly declines. The partial dependence plot for EU industrial production shows a relationship that is strongly negative at the left tail, which indicates that larger fluctuations in industrial production result in a greater chance for a high volatility regime. The plot also confirms the asymmetry that was shown by the EGARCH model: declines in industrial production by about 5 to 15% are strongly associated with the possibility of entering a high volatility regime, but industrial production growth does not reduce volatility risk. The second most important predictor is an EPU shock, which indicates that economic policy uncertainty is able to predict high volatility regimes. The partial dependence plot for the log of the EPU index shows that large, positive EPU shocks increase the log-odds of entering a high volatility regimes which confirms the earlier local projection results.

Rolling Window CV: RF Volatility Regime Classification

Training / Test Window	N Train	N Test	Accuracy	Sensitivity	Specificity	Balanced Acc	RMSE	MAE
2007–2012 / 2013–2014	144	108	0.6204	0.5833	0.6310	0.6071	0.6161	0.3796
2007–2014 / 2015–2016	192	108	0.4537	0.5000	0.4468	0.4734	0.7391	0.5463
2007–2016 / 2017–2018	220	108	0.5000	0.3913	0.5294	0.4604	0.7071	0.5000
2007–2018 / 2019–2020	266	108	0.4630	0.4667	0.4615	0.4641	0.7328	0.5370
2007–2020 / 2021–2022 (Crisis)	**326**	**108**	**0.4259**	**0.5738**	**0.2340**	**0.4039**	**0.7577**	**0.5741**

The rolling window CV shares important information on stability before and after the European energy crisis. The period between 2007 and 2012 shows the best performance with balanced accuracy of 60.7%. The crisis period between 2021 and 2022 exhibits the worst

balanced accuracy of 40.4% which shows how unpredictable this period was. This means the European energy crisis genuinely represents a structural break that the model couldn't have predicted being trained on pre-crisis patterns.

RF Classification Performance: Post-2021 Test Set

Model	N Train	Accuracy	Sensitivity	Specificity	Balanced Accuracy
RF: Downsampled Balanced	**326**	**0.5344**	**0.5392**	**0.5287**	**0.5340**
RF: Unweighted (baseline)	653	0.4603	0.0000	1.0000	0.5000
Naive	NA	0.5873	0.7451	0.4023	0.5737
Majority Class	NA	0.4603	0.0000	1.0000	0.5000

Addressing overall model performance, the balanced Random Forest model achieves 53.4% overall accuracy and 53.4% balanced accuracy, which means that the model correctly identifies 55 out of 102 high volatility regimes and 46 of 87 low volatility regimes. Relative to the naïve benchmark, the balanced Random Forest model has lower overall accuracy: the naïve benchmark also has higher balanced accuracy at 57.4% and higher sensitivity at 74.5%, compared to the Random Forest's model of 53.9% sensitivity. That being said, the naïve benchmark has worse specificity relative to the balanced Random Forest model, which means that it's more likely to overpredict high volatility regimes.

6. Conclusion

The purpose of this paper was to model European household electricity price volatility, and to investigate how the EPU index impacted volatility across 27 European countries in the post-2021 period. The paper employed volatility models such as the GARCH and EGARCH models, local projections, and the Random Forest model in order to predict volatility regimes. The GARCH(1,1) model revealed near-universal volatility persistence across EU member states, and EGARCH(1,1) identified significant asymmetric leverage effects in 14 out of the 27 EU countries. This confirms findings in traditional literature that negative shocks increase electricity price volatility more than positive shocks of the same magnitude. Local projections analysis as outlined by Jorda (2005) was able to pinpoint which horizons exhibited a significant impact on electricity price: horizons 0 to 2 did not show a meaningful impact of EPU shocks on electricity price returns, while horizon 3 displayed a significant negative response and horizons 5 and 6 showed a significant positive response. Lastly, the Random Forest model identified EU industrial production and the EPU index as the two most important predictors in identifying high volatility regimes.

6.1. Contribution to and positioning within existing literature

Traditional literature on European energy markets has tended to focus on commodity markets as well as energy asset volatility which tends to rely on daily or monthly data (Ozdemir 2025, Kim 2026). By contrast, this paper utilized a semi-annual dataset as well as a 27-country panel in order to model the volatility of EU household electricity prices. The application of Jorda's Local Projections analysis (2005) in particular was beneficial for a dataset that is measured semi-annually. This paper can be interpreted as a direct contribution to Ganczarek-Gamrot (2023) who analyzed volatility dynamics in the period after the European energy crisis and how they were

significantly altered. The analysis is extended by using country-level EGARCH leverage estimates across EU member states, including an uncertainty index, and using the Random Forest model to predict the probability of a high volatility regime.

6.2. Methodological limitations and potential remedies

There are a number of methodological limitations to address: the first is the dataset of the dependent variable, which is recorded semi-annually and therefore provides a limited number of observations. While Local Projections analysis can work well with limited datasets, volatility models and machine learning models often require a greater number of observations. A potential solution to this small sample size is the use of a dataset that is measured monthly, or complement the Eurostat dataset with high-frequency wholesale price data. Another limitation is the fact that the Random Forest achieved 53.4% balanced accuracy on the post-2021 test set, which indicates that models that were trained on pre-crisis data have poor predictive capabilities when it comes to potential structural breaks. A possible remedy to this problem is the use of an Adaptive Random Forest model that is able to be retrained sequentially.

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